

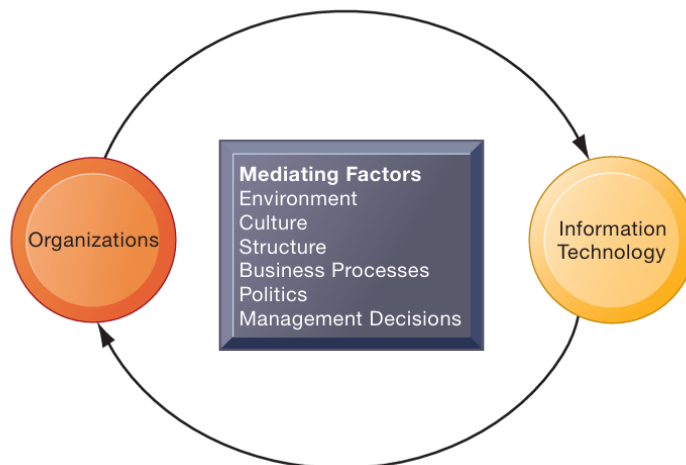
BA 3204 — Fundamentals of MIS
Chapter 3 Information Systems, Organizations, and Strategy

Learning Objectives - By the end of this chapter, you should be able to:

- Explain which features of organizations managers need to understand to build and use information systems successfully.
- Describe the impact of information systems on organizations.
- Explain how Porter's competitive forces model, the value chain model, synergies, core competencies, and network economics help companies build competitive strategy using information systems.
- Identify the challenges posed by strategic information systems and how organizations can manage them.

3.1 Which features of organizations do managers need to know about to build and use information systems successfully?

Information systems and organizations influence each other. Managers build information systems to support business needs, while organizations must also adapt to the changes brought by new technologies.



This relationship is affected by factors such as organizational structure, business processes, culture, politics, external environment, and management decisions. Therefore, managers must understand their organization to design and implement effective information systems. New technology can also create

unexpected effects, such as email and instant messaging becoming major forms of business communication and causing managers to receive hundreds of messages each day.

What Is an Organization?

An organization is a stable, formal social structure that takes resources from the environment and processes them to produce outputs. There are two ways to define an Organization:

- **The Technical definition:** An organization is a formal legal entity that uses inputs such as capital and labor to produce products and services through a production process. It has a structured system of internal rules, procedures, rights, responsibilities, and obligations, and operates within a legal and social environment.
- **The Behavioral definition:** The behavioral view sees an organization as a collection of people and relationships where employees develop customary ways of working and establish agreements about how, how much, and under what conditions work will be done.

These two definitions are not opposites; they complement each other. The technical definition explains how firms combine capital, labor, and technology in a competitive market. The behavioral definition explains what happens inside one specific firm when a new system is introduced. This is why installing new technology is rarely 'just technical.' It touches the way people work, and that is often the harder part to manage.

Common Features of Organizations

Even though every company is different, almost all organizations share certain features. Managers need to understand these features because they directly shape how information systems are designed, accepted, or resisted.

- **Routines and business processes:** Routines (also called standard operating procedures) are the precise rules and practices that employees follow to handle expected situations efficiently. A collection of routines forms a business process, and a collection of business processes forms the firm.
- **Organizational politics:** People in an organization have different roles, interests, and perspectives, which can lead to competition and conflict over resources, rewards, and decision-making. This creates organizational politics and can make change, especially the implementation of new information systems, difficult. Managers who understand and manage these political issues are more likely to successfully implement new systems.

- **Organizational culture:** Organizational culture is the set of shared, deeply held assumptions and beliefs about what the organization should do, how it should work, and who it should serve. It creates common understanding, reduces conflict, and encourages shared practices. However, culture can also resist change, especially when new technology challenges existing beliefs and ways of working. In such cases, the organization may need time for its culture to adapt to the new technology.
- **Organizational environment:** Organizations and their environment have a reciprocal relationship. Organizations depend on the environment for resources, customers, employees, and support, while they can also influence the environment through advertising, partnerships, and other actions. Information systems help organizations monitor external changes such as new technologies, competitors, customer preferences, and government regulations so managers can respond effectively.

However, environments often change faster than organizations, and internal culture, politics, routines, and limited resources can make adaptation difficult.

Disruptive technologies are new technologies or innovations that significantly change markets, industries, or existing business models. They may replace existing products or provide cheaper alternatives that eventually become major competitors. Examples include cars replacing horse-drawn carriages, word processors replacing typewriters, and digital photography replacing film.

Some firms successfully adapt to or create disruptive technologies and gain profits, while others fail because their existing products or business models become obsolete. Being the first to introduce a technology does not always guarantee success; sometimes fast followers benefit more.

- **Organizational structure:** Every organization has a specific structure, and this structure influences the type of information systems it uses and the problems those systems may have. For example, hospitals may have separate systems for doctors, nurses, and administration; small businesses may have quickly developed systems that become inadequate as the business grows; and large multinational firms may have different systems in different locations or divisions without one integrated system. Mintzberg identified five basic structures, summarized below.

Organizational Structure	Description	Example
Entrepreneurial	A young, small firm in a fast-changing environment. Simple structure, run by a single founder/CEO.	A small start-up business

Organizational Structure	Description	Example
Machine bureaucracy	A large bureaucracy in a slowly changing environment, producing standardized products with centralized decision making.	A mid-size manufacturing firm
Divisionalized bureaucracy	A combination of several machine bureaucracies, each making a different product, under one central head office.	General Motors, other Fortune 500 firms
Professional bureaucracy	A knowledge-based organization where output depends on the expertise of professionals; weak central authority.	Law firms, hospitals, universities
Adhocracy	A task-force organization built around short-lived, multi-disciplinary teams responding to rapidly changing environments.	Consulting firms

- **Other features:** Organizations also differ in their goals (coercive, utilitarian, or normative), the groups they serve, their leadership style, and whether their main tasks are routine or non-routine.

3.2 How Do Information Systems Affect Organizations?

Information systems have become central to daily decision making in organizations. Two sets of ideas, one from economics and one from sociology, help explain how and why organizations change once new information technology is introduced.

The Economic View

- **IT substitutes for labor and capital:** As the cost of information technology falls, firms use it in place of other, more expensive resources such as clerical staff, buildings, or machinery. This can reduce the number of middle managers and clerical workers a firm need.
- **Transaction cost theory:** Firms try to reduce the cost of doing business with outside parties (finding suppliers, negotiating, monitoring contracts). Information technology, especially

networks, lowers these transaction costs, which makes it cheaper for firms to outsource work instead of doing everything in-house. This is one reason companies such as Cisco and Dell outsource production instead of owning every factory themselves.

- **Agency theory:** A firm can be seen as a set of contracts between self-interested people: owners (principals) who hire employees (agents) to work on their behalf. Supervising agents is costly. Information technology reduces this supervision cost, allowing managers to oversee more employees with less effort, which again helps shrink the number of middle managers needed.

The Behavioral View

- **IT flattens organizations:** When information is widely shared, lower-level employees get the information they need to make their own decisions without waiting for a supervisor. This pushes decision-making rights down the hierarchy, so fewer layers of middle management are needed, and each manager can supervise a wider span of employees.
- **Postindustrial organizations:** In knowledge-based organizations, authority comes from expertise rather than formal job title. Professionals tend to manage themselves, and decision making becomes more spread out. This also encourages task-force style teams, where specialists come together for a project and then disband once it's finished.

Why Do Organizations Resist New Information Systems?

Information systems often create organizational resistance because they affect access to information, work routines, responsibilities, decision-making, and power. Employees may resist new systems because they require new skills, training, extra effort, or changes in established ways of working. Resistance is influenced by the technology, organizational structure, culture, people, and tasks involved. Therefore, successful implementation requires managing both the technology and the people/organization. Many IT projects fail not because the technology is poor, but because of organizational and political resistance to change.

The Internet's Effect on Organizations

The internet lowers transaction and agency costs even further. It allows companies to distribute manuals, price updates, and instructions to employees and partners almost instantly, and it lets outside vendors check a retailer's internal sales data to plan deliveries. The general pattern is: simpler business processes, fewer employees required for coordination, and flatter organizations.

3.3 Using Information Systems for Competitive Advantage

In every industry, some firms consistently outperform others. Toyota in autos, Amazon in online retail, Google in search. These firms have a competitive advantage: either they have access to resources others do not, or they use common resources more efficiently. Michael Porter developed two well-known models that help explain where competitive advantage comes from, and how information systems can help a firm build it.

Porter's Five Competitive Forces

Porter's model says that a firm's position is shaped not only by its direct competitors, but by five forces acting together:

- **Traditional competitors:** Existing rivals constantly look for new ways to attract your customers through better products, pricing, and branding.
- **New market entrants:** In industries with low barriers to entry (for example, opening a small retail shop), new competitors appear easily. In industries with high capital and knowledge requirements (for example, computer chips), entry is much harder.
- **Substitute products and services:** Customers can switch to an alternative if your price rises too much. For example, streaming music services substitute for CDs, and ride-hailing apps substitute for traditional taxis.
- **Customers' bargaining power:** Customers gain power when it is easy to switch to a competitor, or when prices are fully transparent (as they often are on the internet).
- **Suppliers' bargaining power:** A firm has more control over its suppliers when it has several suppliers to choose from for the same input.

Managers can respond to these five forces using four generic strategies, and information systems play a direct role in each one.

Strategy	What It Means	Example
Low-cost leadership	Use information systems to cut operating costs and offer the lowest prices while still maintaining quality and service.	Walmart's continuous inventory replenishment system

Strategy	What It Means	Example
Product differentiation	Use information systems to create new products, services, or a level of convenience that is hard for rivals to copy, including mass customization for individual customers.	Apple's iPod/iTunes; Nike's custom NIKEiD sneakers
Focus on market niche	Use information systems to analyze customer data closely and serve one narrow segment better than anyone else.	Hilton Hotels' OnQ guest-preference system
Customer and supplier intimacy	Use information systems to build strong, sticky relationships with customers and suppliers, raising the cost of switching to a competitor.	Amazon's purchase recommendations; Chrysler's shared production schedules with suppliers

How the Internet Changed Competition

Because the internet runs on universal standards that any company can use, it has made it easier for new competitors to enter markets and for customers to compare prices instantly. This generally increases rivalry and puts downward pressure on profits. At the same time, the internet has created entirely new markets and business models, from Amazon and Alibaba to YouTube and Airbnb. A good example of this ongoing shift is the Internet of Things (IoT): everyday products such as running shoes, water bottles, and farm tractors are now fitted with sensors that connect them to the internet, turning simple products into 'smart products' that create new services and new sources of customer loyalty.

The Value Chain Model

Porter's five forces model explains where competitive pressure comes from, but it does not tell you exactly where inside your company to act. The value chain model fills this gap. It treats the firm as a chain of activities that each add a bit of value to the final product or service.

- **Primary activities:** Directly related to producing and delivering the product: inbound logistics, operations, outbound logistics, sales and marketing, and service.

- **Support activities:** Make the primary activities possible: firm infrastructure and administration, human resources, technology development, and procurement.

At each stage of this chain, you can ask: how can information systems make this activity more efficient, or improve the relationship with customers and suppliers? This same logic can be extended beyond one company to an entire industry, forming what is called a value web: a network of firms that use information technology to coordinate production and respond quickly to changes in supply and demand, rather than working in a single straight line.

3.4 Challenges of Strategic Information Systems

Building a strategic information system is not a one-time fix. It creates ongoing management challenges that every firm has to plan for.

- **Sustaining competitive advantage:** Competitors can copy or catch up to a strategic system, so the advantage it provides rarely lasts forever. Systems that were once cutting-edge, such as American Airlines' SABRE reservation system or Citibank's ATM network, eventually became something every competitor needed just to stay in business.
- **Aligning IT with business objectives:** Research shows that firms which successfully align their information technology with their business goals are significantly more profitable, yet only a small share of firms actually achieve this alignment. Business leaders need to actively shape how IT is used rather than leaving decisions entirely to technical staff.
- **Managing strategic transitions:** Adopting a strategic system usually requires changes to business goals, relationships with customers and suppliers, and internal processes at the same time. These combined social and technical changes are called strategic transitions, and they are often difficult and slow to complete.

Management Checklist: Questions to Ask Before Investing in a Strategic System

- **Industry structure:** What are the competitive forces in our industry, and how is the industry currently using information technology?
- **Value chain:** Where in our value chain could information systems create more value for the customer or more profit for the firm?
- **Strategic alignment:** Have we clearly defined our business strategy, and is our IT investment actually supporting that strategy?

Review Questions

1. Explain the difference between the technical and behavioral definitions of an organization.
2. List and briefly describe the five basic organizational structures identified by Mintzberg.
3. Why do employees sometimes resist a new information system even when it is meant to make their jobs easier? Use the four resistance factors discussed in class.
4. How does information technology reduce transaction costs and agency costs? Why does this matter for firm size?
5. List Porter's five competitive forces and give one real-world example for each.
6. Name the four generic strategies firms use to respond to competitive forces, and give an example company for each.
7. Distinguish between primary activities and support activities in the value chain model.
8. Why is competitive advantage from information systems often temporary rather than permanent? Support your answer with an example from the chapter.